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Received: 31 January 2023 | Accepted: 24 September 2023

DOI: 10.1002/mar.21919

RESEARCH ARTICLE

Psychology & Marketing WILEY

Is it FOMO or is it ME? The influence of personality traits on cryptocurrency consumption

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Funding information
Southern Illinois University; 2023 College of Business & Analytics Summer Faculty Research Grant

Abstract

Why do consumers invest in cryptocurrency despite its sharp increases and decreases in value? While it seems rational that such volatility should increase consumers' fear of investment, we posit that the fear of missing out (FOMO) on the possible positive returns negates such. Further, this study explores the impact that consumers personality traits, using the big five model, has on FOMO when predicting consumers purchase intentions of cryptocurrencies. Additionally, while consumer behaviorists theorize that information search is an important aspect of the consumer decision making process, marketers have yet to integrate consumers degree of cryptocurrency knowledge as a conditional factor that influences their cryptocurrency purchase behavior. Grounded in trait and prospect theory, we employ a repeated cross-sectional sampling design to investigate these relationships in a bear and bull market. An analysis of the data from 444 and 250 respondents suggests that FOMO predicts the likelihood of investing in cryptocurrencies for certain personality traits. The findings of this study are novel as we observe that FOMO is a driving force behind cryptocurrency's consumption for agreeable and neurotic consumers. However, FOMO negatively mediates cryptocurrency investment likelihood for consumers with high openness to experience and conscientiousness in a bear market, but not in a bull market.

KEYWORDS
big five personality traits, cryptocurrency, FOMO, knowledge, prospect theory

1 | INTRODUCTION

Cryptocurrency is becoming a staple asset in the United States where about 44 million Americans, nearly 17% of the population, are estimated to own cryptocurrency (Faverio & Sidoti, 2023; Kids Count Data Center, 2023). Yet, many of these consumers view cryptocurrency ownership as an investment rather than a replacement for cash (Anaza et al., 2022). These figures are staggering given that it rivals the primary form of investment in the United States, the stock market. Although 54% of Americans hold some form of stock, mostly from investment through retirement accounts, only 14% directly seek to invest in and actively trade in stocks (Ghilarducci, 2020). Thus, it is surprising that these consumers are increasingly drawn to invest in cryptocurrency despite the rapid rise and sharp decline in its value in recent years (Armstrong, 2022).

Indeed, the value of cryptocurrency has had volatile fluctuations. For example, in 2017, the price of Bitcoin rose from $1000 to $20,000 in 1 year, a staggering 1900% increase, but then fell dramatically by 70% to $6000 in 2018 (Yahoo Finance, 2022). The volatility with cryptocurrencies was again showcased when Bitcoin's price rose from $6965.72 in January 2020 to an all-time high of $68,991 (an almost 900% increase), only to fall to $16,000 (an approximate 80% decrease) by December 2022 (Yahoo Finance, 2022). The plummeting price of Bitcoin along with other major cryptocurrencies such as Ethereum, Litecoin, Solana, and Terra Luna has cost investors billions of dollars with many losing their life savings

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(Levy & Sigalos, 2022). This volatility is part of the reason major cryptocurrency exchanges, such as 3AC, Voyager Digital, Celsius, and FTX, filed for bankruptcy (Olinga, 2022).

As with any form of financial investment, consumers invest with the hopes of getting greater future returns. Although this allows consumers to take more calculated risks, no investment is completely risk free. Cryptocurrencies have been shown to carry greater volatility and unprotected risks than other investments such as stocks and gold (Levy & Sigalos, 2022; Olinga, 2022). Yet, many consumers still invest in cryptocurrencies, and for some, the dropping prices seemingly signals a perfect time to buy instead of stoking fears of a crypto bubble burst (Partz, 2022). This then begs the question, what force is motivating the consumption of cryptocurrency as an investment product knowing the risks involved? If these risks are not fear-hindering factors, what fear-motivating factors encourage cryptocurrency consumption? Could consumers be motivated by a fear of missing out (FOMO) that overrides rational evaluations of the risk involved? And if so, what other factors mediate or moderate this relationship? Fear is a powerful negative emotion that influences human cognition, social processes, and behaviors, including investment decisions (Hayes et al., 2022). The evaluation of risks causes investors to rely on fear-based emotions which sometimes lead to poor and wrong investment decisions. As consumers fear that they might miss out on greater return opportunities, FOMO becomes a primal emotion that could influence their decision-making.

FOMO is a powerful psychological motivator since consumers fear being excluded from opportunities when there is a chance for potential returns. The psychological force behind FOMO causes consumers to make decisions without proper investigation or knowledge of the associated risks, which can lead to costly mistakes if the outcome is negative but can also instigate regret if the outcome is positive (Przybylski et al., 2013). Personality traits have recently been used to understand why consumers feel FOMO, as certain personality types make consumers more susceptible to FOMO (Winick, 2021). For example, individuals high in extraversion are more likely to experience FOMO, as these consumers are driven by the desire to socialize and be part of a group (Winick, 2021). Although scholars believe that personality research can inform FOMO (e.g., Winick, 2021), we are unaware of any existing studies that have explored the influence of personality traits on FOMO in the context of cryptocurrency purchase intentions. Additionally, given the novelty of cryptocurrency consumption and the fragmented state of cryptocurrency research in marketing, it is not surprising that no prior study has examined the conditional effect of cryptocurrency knowledge on consumers purchase intentions. Therefore, the purpose of this study is to investigate how personality traits and FOMO influence consumers purchase intentions of cryptocurrencies, while examining the conditional role of consumers cryptocurrency knowledge during a bear versus bull market (see Figure 1).

To this end, this paper is constructed as follows: First, we ground our research in trait and prospect theory to explore the relationship between the big five personality traits and FOMO. Next, we explore the literature on cryptocurrency knowledge. We then formally introduce our model and delineate our methodology and procedure for testing each hypothesis across two studies. Finally, we interpret our results and discuss our findings.

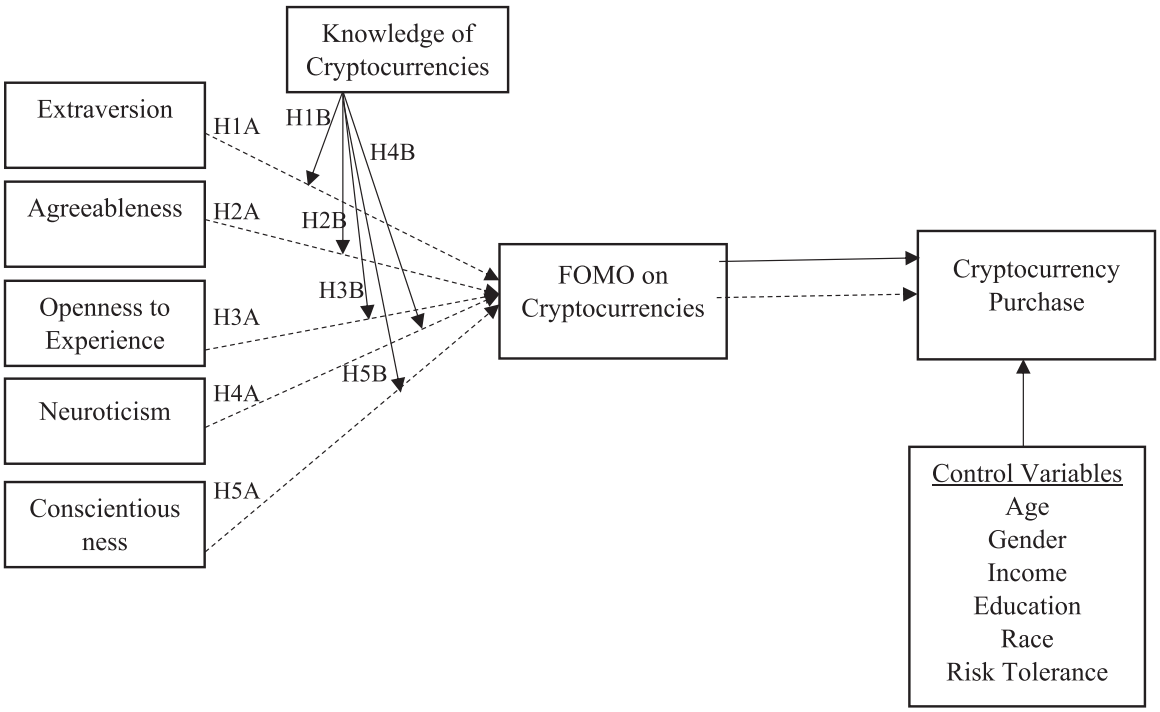


FIGURE 1 Conceptual Model. Dotted lines show the mediating effects of FOMO, solid lines show the moderating mediating effects. FOMO, fear of missing out.

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2 | THEORETICAL DEVELOPMENT

2.1 | Trait theory and the big five personality traits

Trait theory suggests that people's behaviors are based on their personality traits. Personality traits refer to psychological characteristics that make up an individual's unique disposition, along with their way of thinking, feeling, and behaving (American Psychological Association, 2023). These personality traits are often thought to be the result of both nature (innate, genetic) and nurture (environmental influences) (Briley & Tucker-Drob, 2014). In fact, about 50% of the variance in personality traits is construed from a person's hereditary characteristics (genes), while the remaining 50% is deduced from environmental influences such as demographic factors and life experiences (Briley & Tucker-Drob, 2014). Consequently, personality traits are believed to be stable throughout an individual's life, thus providing a better blueprint for understanding people's behavior over time (Briley & Tucker-Drob, 2014).

Decades of research on personality traits have revealed that these traits can predict various kinds of consumer behaviors, preferences, and outcomes (For examples see Digman, 1990). The robustness of personality research by academic, independent, and National Institute of Health (NIH) investigators has led to the systematic testing and validation of a clearly distinguishable taxonomy of five superordinate personality traits, known as the Big-Five (See Digman, 1990 for a review). These big five personality traits are *openness to experience*, *conscientiousness*, *extraversion*, *agreeableness*, and *neuroticism*.

Openness to experience, often associated with creativity and innovative thinking, describes people who accept new ideas, while remaining flexible with their thoughts and actions (McCrae & Costa, 1997). People who are highly open to experience are cognitively curious, visually perceptive, and value leadership roles (McCrae & Costa, 1997). Previous studies have shown that openness to experience is strongly related to variety seeking in consumers' evaluation of products and services but not related to impulsive buying (Olsen et al., 2015).

Conscientiousness is the tendency to be organized and driven, adhering to rules and standards, and striving for excellence in comparison to outside norms (McCrae & Costa, 1997). Highly conscientious individuals are goal driven, responsible, and possess a stronger degree of emotional control (McCrae & Costa, 1997).

Extraversion is a tendency toward being outgoing, enthusiastic, and sociable (McCrae & Costa, 1997). Extroverts prefer human connection and social interaction (Mooradian & Olver, 1996), but can be less empathetic to others in an e-service context (Anaza, 2014). Studies have determined that consumers who are more extroverted tend to enjoy activities that bring enjoyment and pleasure, such as hedonic shopping (Guido, 2006) and online shopping (Bosnjak et al., 2007) because it allows them to discover new products and trends.

Agreeableness refers to the fundamental desire for communal cooperation (McCrae & Costa, 1997). People who are agreeable prioritize getting to know others and are typically courteous, kind, giving, honest, helpful, and willing to sacrifice their own interests for the sake of others (McCrae & Costa, 1997). Consumers with higher levels of agreeableness tend to complain and bargain less while making purchases (Harris & Mowen, 2001; Kowalski, 1996). This is because agreeable consumers are more likely to be satisfied with the products they receive, trusting in the value they paid for that product.

Neuroticism, often known as emotional instability, is the tendency to experience negative emotions such as anger, anxiety, or depression (McCrae & Costa, 1997). A neurotic disposition is associated with a limited tolerance for unpleasant circumstances (Norris et al., 2007) and situations with uncertain outcomes (Olsen et al., 2015). Studies have shown that highly neurotic consumers make more purchases on impulse and are less of variety seekers when shopping due to their inability to postpone gratification (Olsen et al., 2015). These consumers are also more inclined to prefer utilitarian (functional) products over hedonic (experiential) products because of their greater risk-aversion (Guido, 2006).

2.2 | Prospect theory, FOMO, and cryptocurrency

2.2.1 | Prospect theory

Prospect theory (Kahneman & Tversky, 1979) is a psychological theory of behavioral economics and behavioral finance that explains how consumers make decisions under uncertainty. This theory states that consumers tend to be more cautious when faced with the possibility of losses and are more likely to take risks when presented with potential gains. Prospect theory has been used to understand decision making in a variety of contexts, including risk assessment, financial investment, and consumer decision making (Barberis, 2013).

Prospect theory introduces the concept of "reference points," which are the points of comparison that consumers use to evaluate potential outcomes and "corresponds to the current asset position, in which gains and losses coincide" (Kahneman & Tversky, 1979, p. 274). Consumers' investment decisions, as an example, rely on their anticipation for positive returns. Thus, based on their reference points, consumers are more likely to make investments where there are more gains than losses because the pain associated with losing is higher than the satisfaction of gaining. Consequently, consumers' prospect of investing in cryptocurrencies will result only if they perceive that these investments will give them multiple gains and fewer losses.

In our research context, consumers may use reference points to evaluate the potential outcomes of investing in cryptocurrency. For example, if people's reference points are their current financial situation, they may be more likely to invest in cryptocurrency if they perceive it as a potential gain relative to their reference point. This may also lead consumers to avoid investing in cryptocurrency if they perceive it as a potential loss relative to their reference point. Reference points can also be defined relative to other consumers and their financial behaviors such that a consumer's FOMO is formed

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based on other consumers' actions driven by the asset's (cryptocurrency) position and changes in the market.

2.2.2 | FOMO

FOMO is the conscious uneasiness of being uninvolved in a presumed enjoyable or rewarding activity/action experienced by others (Przybylski et al., 2013). FOMO is a self-regulatory dilemma caused by momentary or persistent deficiencies in one's psychological need fulfillment that makes a person feel excluded from an experience (Przybylski et al., 2013). Consumers who experience higher levels of FOMO typically have a need to socialize and interact with others, as well as greater concern for what others are doing to avoid missing out on something presumed to be fulfilling, fun, or important (Przybylski et al., 2013).

FOMO is believed to influence crypto trading since interested consumers may use reference points such as the actions of their peers or marketplace buzz and hype to evaluate the potential outcomes of investing in cryptocurrency (Delfabbro et al., 2021). Here, consumers' prospect of investing may increase when they experience FOMO as FOMO creates the illusion of gains. As the illusion of gains increases, consumers become progressively drawn to invest. FOMO can cause consumers to experience pain especially when a lack of investment is perceived to be a lost opportunity. As the pain of losing is higher than the satisfaction of gaining, FOMO can act as the driving force behind cryptocurrency purchase.

FOMO in cryptocurrency trading can be triggered by a range of factors, one of which is the rise of new entrants that appear to be growing in popularity. Witnessing other people attain favorable results through their investments in these newfound coins can induce a sense of apprehension about not being part of the action, as exemplified by the Dogecoin rally of 2021. Despite Dogecoin having no real underlying value, its price surged by over 10,000% in the first half of 2021 (Gola, 2023), largely due to endorsements from celebrities like Elon Musk and Mark Cuban. Many people, anxious about missing out on potential profits, hurriedly invested in the hype.

Moreover, the influence of social media and online speculation can further intensify FOMO. Platforms like YouTube and social networking sites are filled with discussions and predictions about cryptocurrency prices, fueling consumers' worries of missing out on the next big price movement. For instance, the recent buzz surrounding Shibarium, a blockchain solution developed by the Shiba Inu project, has prompted consumers to speculate about the future price of the Shiba Inu coin, fueling a spike in purchase (Palmer, 2023). Additionally, the desire to seize upon the next major trend and experience massive price oscillations can equally contribute to FOMO in cryptocurrency trading. Those who missed out on previous rallies, like the Dogecoin rally or the soaring price of Bitcoin, may feel compelled to invest hastily in the hope of profiting from future market trends.

Although there is potential for gains, succumbing to FOMO in cryptocurrency trading can result in various adverse consequences. Impulsive trading decisions based on unverified information, especially from self-proclaimed crypto experts, can lead to panic buying and temporary price surges in coins like Shiba Inu and others. However, due to the market's volatility, these panic-induced price increases often lead to massive dumps (massive selling off), causing the value of the cryptocurrency to plummet significantly. As a result, consumers who bought the coins at peak prices suffer substantial financial losses, causing both financial strain and psychological distress. The irrational decision-making caused by FOMO can cause consumers to miss out on potentially profitable moves in the market. For instance, during the bull market of 2021–2022, many individuals invested in numerous cryptocurrencies without proper research. As a consequence, they later discovered that many of those coins had little to no real value in 2023 (Levy & Sigalos, 2022).

Moreover, FOMO can make consumers vulnerable to exit scams, particularly when they fail to conduct thorough research. Coins like "pepe coin" which lack any practical utility attract buyers due to hype and FOMO. The recent spike in pepe coin's price, driven by FOMO, could potentially be a bull trap which may soon lead to a price correction (Chowdhury, 2023). While some traders may profit from the initial pump and dump, others inevitably suffer substantial losses. Furthermore, FOMO can have severe psychological implications for cryptocurrency traders, leading to detrimental emotions such as panic, impatience, jealousy, fear, and hope. These emotional reactions can impair judgment and contribute to impulsive purchase intentions and decisions.

2.2.3 | Cryptocurrency

Cryptocurrency is a digital or virtual form of currency that utilizes cryptographic techniques to ensure the security and confidentiality of monetary transactions (Frankenfield et al., 2022). Cryptocurrencies are based on a decentralized network which implies a lack of regulation by any centralized authoritative bodies such as governments or banking institutions (Frankenfield et al., 2022). Instead, they use blockchain technology to record and verify transactions on a decentralized ledger (Frankenfield et al., 2022). Here, each transaction is recorded as a block, and these blocks are linked together in a chain, creating a secure and transparent record of all transactions. This allows for the secure and decentralized transfer of cryptocurrency without the need for central authority. Due to its lack of exchange rate fees, or delays, cryptocurrency is anticipated to displace government issued money (Anaza et al., 2022). Furthermore, because of decentralization, blockchain provides cryptocurrencies with more security than any other technology currently employed by banks and other financial institutions (Gleim & Stevens, 2021).

Cryptocurrencies[1] can be used to buy goods and services or can be held as investments. But more consumers view cryptocurrencies as an

[1]In 2009, the first cryptocurrency Bitcoin, was created. Since then, other cryptocurrencies have been created and developed, each with their own special qualities and intended applications. Some examples of other popular cryptocurrencies include Ethereum, Litecoin, and Dogecoin.

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investment rather than a currency because they are highly volatile and carry a high level of risk, as their value can fluctuate significantly over time (Anaza et al., 2022). One key difference between cryptocurrencies and traditional fiat currencies is that the value of cryptocurrency is not directly tied to the economic conditions of a particular country or the actions of the central bank (Anaza et al., 2022). Instead, its value is determined by the laws of supply and demand. It is worth noting that, unlike traditional currencies, cryptocurrencies are not insured by governments or other institutions, which means that if consumers lose their cryptocurrency tokens or if they are stolen, it is difficult to regain them (Wengroff, 2021). This is also one of the reasons why consumers view it more as an investment than as a currency to spend.

While about 17% of Americans have purchased cryptocurrencies, there are differences in knowledge and investment among demographics. Research reveals that 78% of males, whereas only 71% of females, are familiar with Bitcoin and men are twice as likely as women to purchase cryptocurrency (de Best, 2022; Perrin, 2021). Additionally, Bitcoin is the preferred choice among Millennials who regard Bitcoin as a more reliable asset than gold (Adams, 2022). Lastly, cryptocurrency knowledge also varies by ethnicity where white Americans (80%) are more knowledgeable about Bitcoin than Hispanics (66%) and African Americans (61%), suggesting that there are disparities in cryptocurrency awareness among different demographics (de Best, 2022).

2.2.4 | Cryptocurrency knowledge

Cryptocurrency knowledge provides consumers with the ability to make informed and rational judgments about cryptocurrency purchase decisions. Investment knowledge is one of the major factors to consider when making investment decisions (Mandell & Klein, 2009). Having financial knowledge is linked to making a greater range of financial decisions, like engaging in the stock market, diversifying a portfolio, and having the ability to stay away from bad investments and subsequently away from debt (Lusardi & Tufano, 2015). Researchers have found that an increase in a person's financial literacy about stocks and the stock market increases his/her likelihood of participating in the stock market (Lusardi & Tufano, 2015). As cryptocurrencies are another form of investment, we believe that cryptocurrency knowledge should increase consumers' awareness leading to increased purchase intentions.

3 | HYPOTHESIS DEVELOPMENT

3.1 | The role of extraversion and FOMO on cryptocurrency purchase intentions

Although extraversion has been linked to short-term investing strategies (Mayfield et al., 2008), the findings have been mixed. For example, Brown and Taylor (2014) found a negative relationship between extraversion and the likelihood of holding shares. On the contrary, Christelis et al. (2010) observed that extroverts exhibited a higher likelihood of holding company stocks. Extraversion has equally been studied as a predictor of FOMO, and like in the case of short-term investing strategies, these studies yielded mixed results. For instance, Nair et al. (2022) found a positive relationship between extraversion and FOMO in young adults aged 18–25. Their reasoning was that highly extroverted consumers are concerned about being excluded from social connections, newer experiences, or important events (Nair et al., 2022). In contrast, Ruyandy and Kartasasmita (2021) studied problematic internet use in young adults aged 18–25 and found that FOMO negatively mediates the relationship between extraversion and this behavior. Despite mixed findings, researchers concur that there is an association between extraversion and FOMO. However, no research exists on the directionality of this relationship in the context of cryptocurrency purchase behavior. Nevertheless, it has been empirically found that FOMO leads to greater purchase intention of Bitcoins (Martin et al., 2022). Thus, in line with this previous research, we infer that FOMO should positively mediate the relationship between extraversion and the tendency to purchase cryptocurrencies given that extroverts are positively influenced by the actions of others, particularly in instances where there is societal hype and buzz like with cryptocurrency consumption. Thus, it is hypothesized that:

> **H1A.** Extraversion will increase FOMO which increases consumers' purchase intentions of cryptocurrency.

Theoretically, cryptocurrency knowledge will strengthen an extrovert's decisions to purchase cryptocurrency because they are more receptive to marketplace intelligence (Turban et al., 2017). Extroverts are known to receive shared knowledge from social networks about new topics (e.g., cryptocurrencies) as a way of staying enlightened on the latest trends and opportunities among their social groups (Chong et al., 2014). Introverts (low extraversion), on the other hand, are less interested in seeking out knowledge about the latest trends (e.g., cryptocurrencies), and are less inclined to experience FOMO since their actions are not heavily driven by their association with other people, social networks, or societal pressures. Accordingly, we hypothesize that:

> **H1B.** More cryptocurrency knowledge will strengthen FOMO for consumers' high on extraversion leading to increased purchase intentions of cryptocurrency.

3.2 | The role of agreeableness and FOMO on cryptocurrency purchase intentions

Agreeable consumers tend to avoid conflicts, trust information offered by others without reservation, and are believed to have a herd mentality when investing (Ahmad, 2020). Previous studies have demonstrated that agreeableness increases FOMO (Hamutoglu et al., 2020). For instance, in a study attempting to discern predictors

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of social media addiction, Turkish scholars identified FOMO regarding updates and developments from networks on social media sites as a catalyst for social media obsession (Hamutoglu et al., 2020). This addiction was observed to be strongest and present only for agreeable individuals and no other personality type.

In the case of investment intentions, current findings on agreeable consumers have been mixed. For example, a study conducted in the Netherlands found no relationship between agreeableness and crypto market participation (Heinen, 2022), meanwhile, a different study conducted in the Pakistani market found a positive relationship between agreeableness and short-term investment intention (Sadiq & Khan, 2019). Research in terms of equity, on the contrary, has found that agreeableness negatively affects stock market investing (Brown & Taylor, 2014). Although these studies provide inconsistent findings, we argue that when agreeable consumers observe that the cryptocurrency market has increased in price, they may experience FOMO for not investing (Delfabbro et al., 2021). Similarly, if they have already invested in some cryptocurrencies, they may feel FOMO for not having invested more. Since agreeable consumers tend to be more trusting and comfortable in their social relationships, they are more likely to experience feelings of missing out if the cryptocurrency market performs well. Consequently this increases their likelihood of purchasing cryptocurrencies to avoid regret. As such, we hypothesize that:

H2A. Agreeableness will increase FOMO which increases consumers' purchase intentions of cryptocurrency.

Agreeable individuals are more inclined to receive information from other people even to a greater degree than extroverts (Anwar, 2017). For this type of consumer, knowledge acquisition is part of the process of formulating social relationships with others in hopes of building trust and a deeper sense of connection (Anwar, 2017). This is particularly the case for financial literacy in that the acquisition of financial knowledge for agreeable consumers increases their investment decisions (Hamza & Arif, 2019). Agreeable consumers, just like extroverts, are more influenced by external sources of information (Anwar, 2017), which has the potential to increase cryptocurrency investments so as not to be excluded from potential market gains. Agreeable consumers are less intuitive in decision making, instead they are dependent decision-makers who rely on the market or other people's advice when making decisions (El Othman et al., 2020). As such, they explore the view of others before deciding on their own decisions (El Othman et al., 2020). Given marketplace headlines about cryptocurrency being "the next big thing" (Marr, 2022), such knowledge will strengthen FOMO for agreeable consumers. As such, we hypothesize that:

H2B. More cryptocurrency knowledge will strengthen FOMO for consumers' high in agreeableness leading to increased purchase intentions of cryptocurrency.

3.3 | The role of openness to experience and FOMO on cryptocurrency purchase intentions

Openness signals a willingness to take greater risks, including increased stock market participation (Brown & Taylor, 2014). Mayfield et al. (2008) pointed out that among younger US consumers, typically in their college years, those with a personality trait receptive to newer experiences were more apt to consider long-term investing.

Thus, given the risky nature of cryptocurrencies coupled with the risk-taking propensity of highly open-minded consumers (Becker et al., 2012), these consumers will experience a greater uneasiness of not being broad-minded enough to invest in an opportunity, subsequently increasing their intention to purchase cryptocurrencies. Since FOMO is a direct predictor of consumers purchase intention of Bitcoins (Martin et al., 2022), we hypothesize that this relationship is also driven by consumers open-mindedness. Thus:

H3A. Openness to experience will increase FOMO which increases consumers' purchase intentions of cryptocurrency.

Open-minded people in general desire more knowledge. Unlike extroverts and agreeable people who are receptive to information from outside sources, open-minded people are information gatherers that collect knowledge from various sources (McCrae & Costa, 1997). While openness to experiences increases risk-taking behavior, the knowledge of the marketplace's volatility and the fear of losses can simultaneously dissuade bad decision-making (Gullone & Moore, 2000). Individuals high on openness value knowledge acquisition and are known for learning (McCrae & Costa, 1997). For these people, their intelligence is characterized by being able to take rational decisions rather than being persuaded by peer pressure or unjustified emotions. More cryptocurrency knowledge for consumers high on openness should provide these individuals with the cognitive ability to make informed decisions on the basis of reasoning and facts rather than on fear-based emotions. Thus, it is hypothesized that:

H3B. More cryptocurrency knowledge will weaken FOMO for consumers' high in openness to experience leading to decreased purchase intentions of cryptocurrency.

3.4 | The role of neuroticism and FOMO on cryptocurrency purchase intentions

Consumers high in neuroticism have a lower likelihood to invest in short-term investments (Mayfield et al., 2008) and when they do, they are likely to hold less risky assets (Oehler et al., 2017). Recently, Heinen (2022) found a negative relationship between neuroticism and cryptocurrency trading. Explanations for this finding rest on claims that consumers high in neuroticism are more susceptible to making poor decisions when faced with choices (Lahey, 2009), which then increases their negative emotions. Higher levels of neuroticism have been linked with more negative emotions such as anxiety, fear,

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and anger, which affect cognitive functioning and predispose neurotic consumers to fear of failure while making self-perceived important decisions (McCrae & Costa, 1997). These consumers show risk aversion tendencies (Niszczota, 2014) and are prone to anxious behaviors when taking risky decisions (Vigil-Colet, 2007).

Previous research has found people with high neuroticism scores tend to show greater signs of missing out which then leads to problematic social media use (Gugushvili et al., 2022; Sindermann et al., 2021). A cross-sectional study conducted in a Chinese university found that FOMO mediated the relationship between neuroticism and problematic WeChat use (WeChat is a popular Chinese app with multifaceted operational systems) (Sindermann et al., 2021). In fact, Sindermann et al. (2021) argued that since components of neuroticism like anxiety and depression have been found to relate positively with FOMO, this may explain their predisposition to problematic WeChat use. Contrary to problematic social media behavior, we argue that given the financial context of this study, consumers high in neuroticism will have less FOMO toward cryptocurrency purchase given their lower predisposition to engage in risky behavior (Oehler et al., 2017). Since cryptocurrency investments are risky (Levy & Sigalos, 2022) and high neurotic consumers are naturally risk averse, these consumers are likely to refrain from the consumption of such currencies. Additionally, since high neurotic consumers are prone to worry when faced with losses (Cheng, 2018a), we predict that these consumers may be adversely impacted by FOMO when faced with cryptocurrency purchase. Thus, we hypothesize that:

H4A. Neuroticism will decrease FOMO which decreases consumers' purchase intentions of cryptocurrency.

Cryptocurrency knowledge should reduce the anxiety and fear associated with purchasing cryptocurrencies for high neurotic consumers. Knowledge of the marketplace should assist with dispelling misconceptions about the cryptocurrency market, thus leading to an increased positive outlook toward cryptocurrency consumption. As these consumers also show compulsive tendencies (Mowen & Spears, 1999), more knowledge should decrease their impulsiveness. Remarkably, high neurotic consumers are less inclined to acknowledge and utilize the suggestions of experts and friends when making investment decisions (Cheng, 2018a). Based on this, we argue that more knowledge should not impact the mediating role of FOMO given that the fear of investment risk for highly neurotic people may be a more intense feeling than the FOMO on buying cryptocurrencies.

H4B. More cryptocurrency knowledge will not affect FOMO for consumers high in neuroticism leading to the same decreased purchase intentions of cryptocurrency.

3.5 | The role of conscientiousness and FOMO on cryptocurrency purchase intentions

Recent research has documented that conscientiousness has no significant impact on cryptocurrency market participation (Heinen, 2022) and is negatively associated with FOMO (Rozgonjuk et al., 2021). This implies that individuals higher on conscientiousness are deliberate and prudent about the outcomes of their actions and behave more rationally than emotionally. Consumers higher in conscientiousness tend to be more mindful about their decisions, and thus are less likely to be driven by impulse, worry, or regret (McCrae & Costa, 1997). They are not impulsive (Mowen & Spears, 1999), and as such they may not easily be swayed by fear-motivators such as FOMO. Brown and Taylor (2014) found that conscientiousness is negatively related to the probability of holding debt, which further affirms that FOMO on the cryptocurrency market may not influence high conscientious consumers toward such investments. Therefore, we argue that highly conscientious consumers would be less willing to formulate cryptocurrency purchase decisions out of FOMO. Accordingly, we hypothesize that:

H5A. Conscientiousness will decrease FOMO which will decrease consumers' purchase intentions of cryptocurrency.

Conscientious individuals are generally detail-oriented, highly driven, and self-motivated (McCrae & Costa, 1997). They typically make financial decisions after careful planning, which entails thorough market research rather than relying on misconceptions (Cheng, 2018b). This cautious planning process allows individuals higher on conscientiousness to be more selective of their actions (McCrae & Costa, 1997). More knowledge should help highly conscientious consumers gain better insights about cryptocurrencies. Thus, their purchase intention of cryptocurrencies will depend upon the outlook they have formed toward cryptocurrencies from the marketplace and expert opinions. For example, consumers' outlook for Bitcoin as compared to other cryptocurrencies is more positive, but their outlook on the cryptocurrency market is generally negative (Browne & Kharpal, 2023). Given the general negative sentiment toward the cryptocurrency market, higher knowledge should discourage highly conscientious consumers from buying cryptocurrencies. Thus, it is hypothesized that:

H5B. More cryptocurrency knowledge will further weaken FOMO for consumers' high in conscientiousness leading to more decreased purchase intentions of cryptocurrency.

4 | STUDY 1

4.1 | Data collection

The main goal of study 1 is to test hypothesis 1A through 5B at a time when the market is bear in direct response to the price depression, economic decline, and recession in the United States. Amazon Mechanical Turk (Mturk) was utilized for data collection. Mturk was used for data collection for three reasons. First, given consumers' mainstream ability to buy cryptocurrencies through cryptocurrency exchanges such as Coinbase as well as online brokerages like Robinhood, it was important to obtain a representative sample of US consumers and Mturk provided an opportunity to obtain such a

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pool of participants. Second, the timeline of the data collection was critical and Mturk data can be collected in a short period of time (Follmer et al., 2017). The data was collected in August 2022 and during this time, the price of Bitcoin had plummeted by 52% from $47,686.81 on January 1, 2022, to $22.846.51 on August 3, 2022 (Yahoo Finance, 2022). The reason for this massive drop can be attributed to the collapse of the stable coin terraUSD in May 2022 followed by the collapse and bankruptcy of the hedge fund, Three Rows Capital, which had significant investment in terraUSD (Kharpal & Browne, 2022). It was estimated that around $42 billion was lost by the holders of terraUSD and Terra Luna because of this crash (Browne, 2022). Thus, when the respondents were taking this survey, cryptocurrency was in the bear market. Third, the US currently represents the largest market adopting cryptocurrencies and blockchain based platforms, and has the largest number of cryptocurrency holders, followed by India, Pakistan, Nigeria, and Vietnam (Triple A, 2023). Thus, it was imperative that we study the largest cryptocurrency market in terms of valuations.

The survey was launched as part of three separate studies. The survey for this study was used as a distraction for an unrelated short experiment on racial disparity in the United States. Study 1 was associated with the experiment, Study 2 was associated with the current research, and Study 3 asked demographic questions. The survey was distributed to US participants who had a greater than 95% HIT approval rate. Each person who responded was compensated with a small sum of money with each completed response. Participants had to be at least 19-year-old to participate in the study. In line with Kees et al. (2017a.) only respondents registered as residents of the United States were invited to take the survey and protocols were setup to restrict respondents from taking the survey more than once. Additionally, there was one attention check in the survey to ensure participants' concentration. Those participants who failed the attention check were excluded from the study. Together, the geographic and single survey restrictions and the attention check ensures that the data's validity and reliability is equal to, if not greater than, student-generated data or other panel data (see Kees et al., 2017a, 2017b) and have become acceptable standards within marketing research (for examples see Bennett et al., 2022; Yun et al., 2019). Participants were asked to state their responses about the Big Five personality Traits, FOMO, purchase intention, knowledge, and risk tolerance level. Finally, respondents completed the demographic questions regarding their age, gender, income, education, and race. Of the 450 participants, six failed the attention check question, leaving a final sample of 444 participants.

4.2 | Measures

The items for the study were directly drawn or adapted from previous research. The independent variables in our study are the big five personality traits, FOMO is the mediator, knowledge of cryptocurrencies is the moderator, and purchase intention of cryptocurrency is the dependent variable. The items for the big five personality traits (*agreeableness*, *openness to experience*, *conscientiousness*, *neuroticism*, and *extraversion)* were taken from Rammstedt and John (2007). Each personality trait was measured using two items, one of each being reverse scored to reflect participants low on that trait. Participants were asked to rate themselves on 10 items using a five-point Likert scale (1 = *strongly disagree* to 5 = *strongly agree*). The items for FOMO were adapted from Good and Hyman (2021). There were three items for FOMO measured on a five-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). The items for cryptocurrency knowledge were adapted from Algesheimer et al. (2005). Two items, each measured on a five-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*), asked participants to specifically rate the extent of their knowledge about cryptocurrency. The first item reads as follows: "when compared to other people, I know a lot about cryptocurrency." The second asks, "my friends consider me an expert regarding cryptocurrency." Although both questions were asked during data collection, the second was not utilized for the analysis as the opinion of friends were considered second-hand information likely to bias the results. The items for purchase intentions were adapted from Han et al. (2017). Three items measured on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*) asked the participants their intent to purchase cryptocurrencies as a direct indicator of consumer behavior. Table 1 contains the constructs along with their measurement.

4.3 | Control variables

Age, income, gender, education, and race were the demographic variables controlled for in the study firstly to elucidate the explanatory power of FOMO and purchase intentions of cryptocurrency and secondly to ensure that any observed differences in our analysis are not because of demographic factors. Risk tolerance was also controlled for given the causal effect of risk perceptions on financial behaviors (Corter & Chen, 2006). Risk perception was measured in terms of participants risk tolerance level, whether they are conservative or aggressive in taking risks (see Table 1). The item to measure risk tolerance was taken from Ameriprise Financial which is a financial service company assisting consumers in managing their investments such as wealth and assets (Ameriprise Financial, 2023).

4.4 | Sample description

About 86% (86.3%) of the 444 participants were Caucasian, 4.5% African American, 7.9% Asian, 1.1% Hispanic, and 0.2% of other races. The average age of the participants was 36.5 years. Out of the 444 participants, 243 were male (54.7%), 200 were female (45%), and one participant did not prefer to respond. Around three percent (3.4%) earned less than $25,000, 28.6% earned between $25,000 and $49,999, 39.4% earned between $50,000 and $74,999, 20% earned between $75,000 and $99,999, and 8.6% earned $100,000 or more annually. In terms of highest degree earned, 5.6% of the

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TABLE 1 Constructs and items.

Constructs	Items
Big five personality traits (Rammstedt & John, 2007) Item # for each personality trait: Extraversion (1, 6) Agreeableness (2, 7) Conscientiousness (3, 8) Neuroticism (4, 9) Openness to experience (5, 10)	I see myself as someone who: 1. Is reserved (reverse scored) 2. Is generally trusting (reverse scored) 3. Tends to be lazy (reverse scored) 4. Is relaxed, handles stress well (reverse scored) 5. Has a few artistic interests (reverse scored) 6. Is outgoing sociable 7. Tends to find fault in others 8. Does a thorough job 9. Gets nervous easily 10. Has an active imagination
FOMO (Good & Hyman, 2021)	If I do not buy cryptocurrencies: 1. I am afraid later I will feel sorry I did not buy it with my friends 2. I will worry about what I am missing 3. I will worry my friends are doing more rewarding things than me
Knowledge of cryptocurrency (Algesheimer et al. (2005)	1. When compared to other people, I know a lot about cryptocurrency 2. My friends consider me an expert regarding cryptocurrency[a]
Cryptocurrency purchase intention (Han et al., 2017)	1. I intend to buy cryptocurrencies 2. I am willing to buy cryptocurrencies 3. I will make an effort to buy cryptocurrencies
Risk tolerance	Which statement describes you perfectly? 1. I need to see at least a little return 2. I have a hard time tolerating losses 3. I can tolerate a small loss 4. I can tolerate loss 5. I don't mind if I lose money

[a]Not included for data analysis.

respondents had completed high school, 21.8% had completed some college or an associate degree, 47.1% had completed an undergraduate degree, 23% had completed a master's degree, and 2.5% had completed a doctoral or professional degree. On the risk tolerance survey question, 11.9% of respondents indicated that they needed to witness at least a little return before making risky decisions, 18.5% indicated that they had a hard time tolerating losses, 37.2% indicated that they can tolerate a small loss, 20.9% indicated that they could tolerate a loss, and 11.5% indicated that they did not mind losing their money.

5 | DATA ANALYSIS AND RESULTS

Table 2 provides the inter-item correlations among the constructs. As can be seen from Table 2, moderate correlation exists among the constructs. Bivariate correlation analysis shows that FOMO is significantly and positively associated with agreeableness ($r = 0.23$)and neuroticism ($r = 0.26$) but negatively associated with conscientiousness (−0.38) and openness to experience (−0.17). The analysis also showed that cryptocurrency purchase intention was significantly and positively correlated with extraversion ($r = 0.21$). In contrast, consumers purchase intention of cryptocurrency was minimally and negatively associated with agreeableness (−0.13) and neuroticism (−0.10).

To test the hypotheses, we used PROCESS Macro by Hayes (Hayes, 2017). PROCESS Macro is typically employed in empirical analysis to estimate the direct and indirect effects of relationships as well as to assess the conditional role of moderator variables. PROCESS Macro can estimate indirect effects using as much as 5000-bias corrected bootstrap samples (Hayes, 2017). For this study, Model 4 was used to test the mediating relationship of FOMO in explaining the impact of each personality trait on consumers cryptocurrency purchase intention: H1A, H2A, H3A, H4A, and H5A. Thereafter, Model 7 was used to test the moderating effect of cryptocurrency knowledge on the mediating relationship of FOMO when investigating the link from consumers personality traits to their purchase intention of cryptocurrency: H1B, H2B, H3B, H4B, and H5B.

5.1 | Mediation analysis: Direct and indirect effect

H1A argues that extraversion increases FOMO, which then leads to increased purchase intentions of cryptocurrency. Results indicate that the mediating role of FOMO was not supported as the indirect relationship was not significant ($b = 0.004$, 95% CI [−0.032, 0.050]). Thus, H1A is not supported. The analysis also revealed that the effect of extraversion on purchase intention was positive and significant

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TABLE 2 Inter-item correlations, means, and standard deviations.

Study 1										Study 2								
Variable	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
EXT (1)	-									-								
AGG (2)	−0.12**	-								−0.16**	-							
CON (3)	0.05	−0.38**	-							0.21**	−0.30**	-						
NEU (4)	−0.22**	0.30**	−0.33**	-						−0.06	0.61**	−0.29**	-					
OPE (5)	0.05	−0.08	0.17**	−0.09*	-					−0.03	−0.02	−0.04	−0.01	-				
PI (6)	0.21**	−0.13**	0.06	−0.10*	−0.08	-				0.09	−0.16**	−0.07	−0.13*	−0.21**	-			
RISK (7)	−0.06	−0.05	0.02	−0.01	0.05	−0.05	-			0.01	0.01	0.03	−0.02	0.01	0.004	-		
FOMO (8)	0.01	0.23**	−0.38**	0.26**	−0.17**	0.32**	0.009	-		0.06	0.33**	−0.12*	0.36**	−0.09	0.42**	0.01	-	
KNO (9)	0.04	−0.03	−0.02	0.02	−0.04	0.55**	−0.01	0.29**	-	−0.06	−0.11	−0.05	−0.09	−0.06	0.61**	−0.02	0.31**	-
Mean	3.09	2.77	3.28	2.76	3.13	3.97	2.98	3.53	3.76	3.02	2.87	3.15	2.82	3.04	4.10	3.14	3.82	4.02
SD	0.63	0.69	0.73	0.72	0.62	0.63	1.15	0.92	0.84	0.35	0.51	0.49	0.52	0.38	0.58	1.07	0.83	0.62

Abbreviations: AGG, agreeableness; CON, conscientiousness; EXT, extraversion; FOMO, fear of missing out; KNO, knowledge; NEU, neuroticism; OPE, openness to experience; PI, purchase intention; RISK, risk tolerance.

*$p < 0.05$; **$p < 0.01$.

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($b = 0.206$, $p < 0.001$), meanwhile its effect on FOMO was not significant ($b = 0.018$, $p = 0.788$).

As predicted in H2A, we found that agreeableness increases FOMO leading to an increased purchase intention of cryptocurrency as the indirect relationship was significant ($b = 0.083$, 95% CI [0.039, 0.131]). Thus, H2A is supported. The results also demonstrate that agreeableness directly decreases consumers purchase intention of cryptocurrency ($b = -0.211$, $p < 0.001$), while increasing FOMO ($b = 0.323$, $p < 0.001$), further supporting the indirect effect relationship.

H3A hypothesized that openness to experience increases FOMO leading to increased consumers purchase intentions of cryptocurrency. However, the mediating role of FOMO was not supported in the direction it was hypothesized. Interestingly, we find that the mediating relationship is negative and significant ($b = -0.055$, 95% CI [−0.099, −0.013]). Thus, H3A is partially supported given the significant negative effect. Moreover, we found that the direct effect of openness to experience on purchase intention is insignificant ($b = -0.019$, $p = 0.680$). However, the effect of openness to experience on FOMO is negative and significant ($b = -0.255$, $p < 0.001$).

H4A predicted that neuroticism will decrease FOMO which decreases consumers' purchase intentions of cryptocurrencies. The results revealed the effect to be in reverse of the hypothesized directionality. As such, FOMO positively mediated the relationship between neuroticism and purchase intentions of cryptocurrencies ($b = 0.093$, 95% CI [0.053, 0.139]). Thus, while statistical significance is found to show that neuroticism affects FOMO in a cryptocurrency purchase context, the effect is positive, providing partial support for H4A. From the analysis, neuroticism increased consumers' FOMO ($b = 0.357$, $p < 0.001$), but directly decreased consumers purchase intentions of cryptocurrency ($b = -0.191$, $p < 0.001$).

H5A predicted that conscientiousness will decrease FOMO which will decrease consumers' purchase intention of cryptocurrency. As predicted, we found that the indirect effect of FOMO on conscientiousness and purchase intention is negative and significant ($b = -0.134$, 95% CI [−0.193, −0.083]). Thus, H5A is supported. The direct effect of conscientiousness on purchase intention is positive and significant ($b = 0.196$, $p < 0.001$) and more importantly, the effect of conscientiousness on FOMO is found to be negative and significant ($b = -0.482$, $p < 0.001$). Refer to Table 3 for these results.

5.2 | Conditional effects: Moderated mediation analysis

H1B predicted that more cryptocurrency knowledge will strengthen FOMO for consumers' high on extraversion leading to increased purchase intentions of cryptocurrency. Contrary to the hypothesis, the analysis revealed that more cryptocurrency knowledge decreased FOMO for consumers high on extraversion ($b = -0.198$, $p < 0.001$, see Table 4), which did not impact consumers purchase intentions for cryptocurrency ($b = -0.020$, 95% CI [−0.059, 0.024], $p > 0.05$). Consequently, while we find statistical support for the interaction effect on FOMO, the results of H1B are not fully supported for consumers purchase intention. H2B predicted that more cryptocurrency knowledge will strengthen FOMO for consumers' high in agreeableness leading to increased purchase intentions of cryptocurrency. From the moderated mediated analysis, we found that the interaction of agreeableness and knowledge on FOMO was not significant ($b = 0.085$, $p = 0.116$). Nevertheless, the results showed that higher knowledge leads to significant increase in purchase intention of cryptocurrencies for agreeable people through the indirect effect of FOMO ($b = 0.093$, 95% CI [0.044, 0.147]). Thus, H2B was supported.

We predicted in H3B that more cryptocurrency knowledge would weaken FOMO for consumers' high in openness to experience leading to decreased purchase intentions of cryptocurrency. The interaction of openness to experience and knowledge on FOMO was negative and significant ($b = -0.138$, $p < 0.05$). Moreover, the conditional indirect effect also shows that higher cryptocurrency knowledge for open consumers decreases their purchase intentions ($b = -0.067$, 95% CI [−0.118, −0.021]). Thus, H3B was supported.

H4B predicted that more cryptocurrency knowledge will not affect FOMO for consumers high in neuroticism leading to the same decreased purchase intentions of cryptocurrency. However, the interaction of neuroticism and knowledge on FOMO was positive and

TABLE 3 Study 1—Mediation analysis.

Indirect effect			Direct effect			
IV→M→DV	Effect	95% CI	IV→DV	Effect	IV→Med	Effect
EXT→FOMO→PI	0.004	[−0.032, 0.050]	EXT→PI	0.206***	EXT→FOMO	0.018
AGG→FOMO→PI	0.083*	[0.039, 0.131]	AGG→PI	−0.211***	AGG→FOMO	0.323***
OPE→FOMO→PI	−0.055*	[−0.099, −0.013]	OPE→PI	−0.019	OPE→FOMO	−0.255***
NEU→FOMO→PI	0.093*	[0.053, 0.139]	NEU→PI	−0.191***	NEU→FOMO	0.357***
CON→FOMO→PI	−0.134*	[−0.193, −0.083]	CON→PI	0.196***	CON→FOMO	−0.482***

Abbreviations: AGG, agreeableness; CON, conscientiousness; DV, dependent variable; EXT, extraversion; FOMO, fear of missing out; IV, independent variable; M, mediating variable; NEU, neuroticism; OPE, openness to experience; PI, purchase intention.

*$p < 0.05$; ***p < 0.001.

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TABLE 4 Study 1—Moderated mediation analysis.

IV	Med	W	DV	IV* Mod	Conditional effect of high knowledge	95% CI
EXT	FOMO	KNO	PI	−0.198***	−0.020	[−0.059, 0.024]
AGG	FOMO	KNO	PI	0.085	0.093*	[0.044, 0.147]
OPE	FOMO	KNO	PI	−0.138*	−0.067*	[−0.118, −0.021]
NEU	FOMO	KNO	PI	0.203***	0.107*	[0.062, 0.159]
CON	FOMO	KNO	PI	−0.105*	−0.144*	[−0.202, −0.092]

Abbreviations: AGG, agreeableness; CON, conscientiousness; DV, dependent variable; EXT, extraversion; FOMO, fear of missing out; IV, independent variable; KNO, knowledge; M, mediating variable; NEU, neuroticism; OPE, openness to experience; PI, purchase intention; W, moderating variable.
*$p < 0.05$; ***$p < 0.001$.

significant ($b = 0.203$, $p < 0.001$), thus validating the impact of knowledge among neurotic consumers. Moreover, the conditional indirect effect shows that more cryptocurrency knowledge for neurotic consumers leads to higher purchase intentions of cryptocurrencies ($b = 0.107$, 95% CI [0.062, 0.159]), failing to support H4B, but providing novel insights.

Finally, H5B proposed that more cryptocurrency knowledge will further weaken FOMO for consumers' high in conscientiousness leading to more decreased purchase intentions of cryptocurrency. The interaction of conscientiousness and knowledge on FOMO was negative and significant ($b = =-0.105$, $p < 0.05$). Moreover, the conditional indirect effect shows that more cryptocurrency knowledge decreased FOMO leading to decreased purchase intentions of cryptocurrency ($b = -0.144$, 95% CI [−0.202, −0.092]) for conscientious consumers. Thus, H5B was supported.

5.3 | Discussion

This study embarks on a comprehensive evaluation of the role of consumers personality traits and FOMO in a cryptocurrency context. Specifically, the relationship between the Big five personality model and FOMO on cryptocurrency purchase intentions is examined while considering the influences of cryptocurrency knowledge. Although, a growing body of researchers continue to advocate for an in-depth examination of marketing's role in understanding consumers adoption of cryptocurrencies (Anaza et al., 2022), such opportunities cannot be advanced if the psychological motivations behind these decisions and behaviors are not investigated. While these results advance our understanding of the personality traits behind cryptocurrency consumption in a bear market, Study 2 replicates this examination in a bull market to discern the generalizability of these findings.

5.4 | Study 2: Replication in a bull market

The current scenario in August 2023 presents a significant contrast from the scenario in Study 1, as the market has shifted into a bullish phase, witnessing a notable surge in cryptocurrency prices compared to 2022. This change is evident in the closing price of Bitcoin, which increased from $20,049.76 on August 1, 2022, to $26,162.37 on August 1, 2023 (23% increase), as reported by Yahoo Finance.[2] Given this altered market outlook, it becomes imperative to investigate whether consumer sentiments have undergone a transformation in relation to the changing market dynamics.

5.5 | Data collection and measures

To examine potential shifts in consumer sentiment toward cryptocurrencies within a bullish market context, we conducted a replication of study 1 using Mturk. The procedures and compensation mirrored the initial study to enable direct comparison. The inclusion criteria remained the same, with participants required to be based in the United States and have a Mturk approval rating above 95%. The same measures utilized in study 1 were employed in this study. To capture market perceptions, two additional questions asked participants to reflect on their outlook toward the cryptocurrency market 1 year prior as well as 1 year from the point of data collection. In total, 255 participants completed the survey. Five responses were removed for failing an attention check question or leaving responses blank. The final sample consisted of 250 complete responses.

5.6 | Sample description

Ninety-three percent of participants identified as Caucasian (93.2%), 0.8% as African American (0.8%), 2% as Asians, and 4% as American Indian or Alaskan Native. The average age was 35.3-year-old. Seventy-four percent of participants (74.4%) identified as male ($n = 186$) while females represented the remaining 25.6% ($n = 64$). In terms of income, 4% earned less than $10,000 annually. Approximately 7.2% made between $25,000–$34,999, 25.6% earned

[2]Study 2's data was collected on September 5 to ensure the 1-year time lag.

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$35,000–$49,999, 28.8% made $50,000–$74,999, 28.8% earned $75,000–$99,999, and 5.6% reported income levels above $100,000. Regarding education, 18.4% completed high school, 8.8% attained some college, 7.2% held associate degrees, 38.8% had bachelor's degrees, 24.4% attained graduate degrees, and 2.4% held doctoral or professional degrees. Eleven percent of participants (11.2%) required minimal returns before making risky decisions. About 21% (20.8%) had difficulty tolerating losses, 48.8% tolerated small losses, 8.8% tolerated losses, and 10.4% did not mind losing money.

5.7 | Data analysis and results

5.7.1 | Mediation analysis: Direct and indirect effect

Similar to the analysis performed in study 1, PROCESS Macro model 4 was used to test the mediation analysis (see Table 5), while model 7 was used to test the moderated mediation (see Table 6). H1A-R (R stands for replication) was not supported seeing that FOMO did not mediate extraversion and purchase intention (b = 0.040, 95% CI [−0.073, 0.179]. Direct effects of extraversion on FOMO ($b = 0.132$, $p = 0.367$) and purchase intention ($b = 0.124$, $p = 0.190$) were also not significant.

H2A-R was supported as FOMO mediated agreeableness and purchase intention ($b = 0.218$, 95% CI [0.089, 0.348]. Agreeableness directly increased FOMO ($b = 0.555$, $p < 0.001$) and decreased purchase intention ($b = -0.397$, $p < 0.001$). H3A-R was not supported since FOMO did not mediate the relationship between openness to experience and purchase intention ($b = -0.064$, 95% CI [−0.200, 0.028]). However, the direct effect of openness to experience on purchase intention was negative and significant ($b = -0.269$, $p < 0.01$). Consistent with findings in study 1 when testing H4A-R, FOMO positively mediated neuroticism and purchase intention ($b = 0.222$, 95% CI [0.087, 0.391]. H5A-R was not supported in that FOMO did not mediate the relationship between conscientiousness ($b = -0.056$, 95% CI [−0.172, 0.019]) and purchase intention. Conscientiousness was not directly related to FOMO ($b = -0.187$, $p = 0.085$) and purchase intention ($b = -0.056$, $p = 0.428$).

5.8 | Moderated mediation analysis

Consistent with findings from study 1, the moderated impact of consumer knowledge did not impact purchase intention via FOMO for consumers high on extraversion ($b = -0.038$, 95% CI [−0.249, 0.067]), failing to support H1B-R. Meanwhile, the interactive effect between extraversion and customer knowledge on FOMO was not significant ($b = -0.248$, $p = 0.149$). These findings also supported H2B-R, consistent with the findings from study 1. There was a significant interaction between agreeableness and knowledge on FOMO ($b = 0.641$, $p < 0.001$). Also, high knowledge enhanced the mediating effect of FOMO, thus increasing purchase intention ($b = 0.420$, 95% CI [0.191, 0.661]). However, Hypothesis 3B-R was not supported. The interaction between openness and knowledge on FOMO was not significant ($b = 0.194$, $p = 0.344$). Furthermore, high knowledge did not influence purchase intention through FOMO's mediating effect ($b = 0.006$, 95% CI [−0.112, 0.163]) for consumers

TABLE 6 Study 2—Moderated mediation analysis.

IV	Med	W	DV	IV* Mod	Conditional effect of high knowledge	95% CI
EXT	FOMO	KNO	PI	−0.248	−0.038	[−0.249, 0.067]
AGG	FOMO	KNO	PI	0.641***	0.420*	[0.191, 0.661]
OPE	FOMO	KNO	PI	0.194	0.006	[−0.112, 0.163]
NEU	FOMO	KNO	PI	0.406***	0.310*	[0.136, 0.510]
CON	FOMO	KNO	PI	0.150	−0.019	[−0.151, 0.068]

Abbreviations: AGG, agreeableness; CON, conscientiousness; DV, dependent variable; EXT, extraversion; FOMO, fear of missing out; IV, independent variable; KNO, knowledge; M, mediating variable; NEU, neuroticism; OPE, openness to experience; PI, purchase intention; W, moderating variable.

*$p < 0.05$; ***$p < 0.001$.

TABLE 5 Study 2—Mediation analysis.

Indirect effect			Direct effect			
IV→M→DV	Effect	95% CI	IV→DV	Effect	IV→Med	Effect
EXT→FOMO→PI	0.040	[−0.073, 0.179]	EXT→PI	0.124	EXT→FOMO	0.132
AGG→FOMO→PI	0.218*	[0.089, 0.348]	AGG→PI	−0.397***	AGG→FOMO	0.555***
OPE→FOMO→PI	−0.064	[−0.200, 0.028]	OPE→PI	−0.269**	OPE→FOMO	−0.219
NEU→FOMO→PI	0.222*	[0.087, 0.391]	NEU→PI	−0.364***	NEU→FOMO	0.565***
CON→FOMO→PI	−0.056	[−0.172, 0.019]	CON→PI	−0.056	CON→FOMO	−0.187

Abbreviations: AGG, agreeableness; CON, conscientiousness; DV, dependent variable; EXT, extraversion; FOMO, fear of missing out; IV, independent variable; M, mediating variable; NEU, neuroticism; OPE, openness to experience; PI, purchase intention.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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high in openness. The interaction between neuroticism and knowledge on FOMO was significant ($b = 0.406$, $p < 0.001$). High knowledge increased purchase intention through FOMO's mediating effect ($b = 0.310$, 95% CI [0.136, 0.510]). As such, H4B-R was not supported, but the results compare to the findings from study 1. Finally, H5B-R was not supported as the interaction between conscientiousness and knowledge on FOMO was not significant ($b = 0.150$, $p = 0.315$). High cryptocurrency knowledge did not impact purchase intention through FOMO's mediating effect ($b = -0.019$, 95% CI [−0.151, 0.068]).

5.9 | Discussion

Study 2 reveals important insights concerning the relationship between personality traits and cryptocurrency consumption during a bull market. Generally speaking, consumers high in extraversion, agreeableness and neuroticism react similarly in a bull and bear market to fears of missing out about cryptocurrency, which subsequently affects their positive or negative decision to purchase cryptocurrency. On the contrary, we observed clear distinctions among highly conscientious and open consumers in a bear and bull market. Study 2 provides initial evidence to demonstrate that highly conscientious and open individuals respond indifferently to their decision to purchase cryptocurrency in a bull market. We particularly detected that during a bull market, highly open and conscientious consumers were uninfluenced by FOMO, which was not the case in the bear market. In the bear market, consumers high on these two personality traits were negatively influenced by FOMO in deterrence of purchasing cryptocurrencies. A possible explanation for these findings could be the role that cryptocurrency knowledge plays during a bear and bull market. As shown in Table 7, cryptocurrency knowledge increased overall for the five personality traits. Notably, we observe that for highly open and conscientious consumers, less knowledge prevented cryptocurrency purchases (refer to study 1), while more knowledge led to purchase indifference. This indicates that highly open and conscientious consumers need greater certainty in the cryptocurrency market before making purchase decisions.

5.10 | General discussion

Study 1 provided valuable insights into consumer FOMO and cryptocurrency purchase intentions during challenging market conditions. However with Study 2, we uncover whether the ongoing bull market has sparked renewed interest and openness among ordinary consumers toward cryptocurrencies, or if cautiousness still prevails despite the price surge. Although cryptocurrency prices have surged, it is evident that mainstream acceptance of cryptocurrencies remains a gradual process. This study illuminates the extent to which the general public has embraced this evolving financial landscape. By scrutinizing consumer personalities and fears in a bear and bull market, we show greater mainstream adoption among certain personality traits, whereas reservations exist among other groups of consumers. To understand why consumers purchase cryptocurrencies, researchers must first comprehend who these consumers are. Using prospect and trait theory, this study identifies the psychological characteristics of cryptocurrency consumption followed by a substantive inquisition into how FOMO and knowledge influence cryptocurrency purchase intentions.

TABLE 7 Cryptocurrency knowledge breakdown.

Personality	Knowledge (mean) study 1	Knowledge (mean) study 2
Low EXT	3.82	4.08
High EXT	3.67	3.68
Low AGG	3.75	4.03
High AGG	3.78	4.01
Low OPE	3.82	4.04
High OPE	3.67	3.92
Low NEU	3.68	3.96
High NEU	3.82	4.04
Low CON	3.78	4.07
High CON	3.74	3.90

Abbreviations: AGG, agreeableness; CON,conscientiousness; EXT, extraversion; NEU, neuroticism; OPE, openness to experience.

5.11 | Theoretical implications

Theoretically, our results confirm trait theory's application as well as prospect's theory relevance to cryptocurrency purchase intention. When seeking to explain drivers of cryptocurrency consumption, the results are extremely insightful. As the article's title suggests: "Is it FOMO or is it ME," we found that the answer to this question depends on the consumer. For consumers who are open to experiences and highly conscientious, FOMO decreased their likelihood of purchasing cryptocurrencies, whereas for agreeable and neurotic consumers, FOMO increased their purchase intentions. Without the presence of FOMO, agreeable and neurotic consumers were not inclined to engage in cryptocurrency participation. In fact, our results demonstrate that the more neurotic and agreeable tendencies a person shows, the lower their cryptocurrency purchase intentions. Contrary to existing research (Nair et al., 2022), extroverts did not appear to be motivated by FOMO but instead were directly willing to purchase cryptocurrencies without any social fear pressures of exclusionism.

These findings extend the existing body of research on the factors influencing cryptocurrency consumption. Given the wide-acceptance and prevalence of the Five Factor Model, it is surprising that no study till date has attempted to understand its relationship with FOMO within the context of cryptocurrency consumption.

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Martin et al. (2022) identified the impact of the dark tetrad of personality traits on Bitcoin purchasing intentions controlling for FOMO as a mediator. Our work extends Martin et al.'s research by identifying the mediating role of FOMO on a generally accepted taxonomy of human personality traits and consumers likelihood of buying cryptocurrency. These findings conclusively show that FOMO shockingly increases the tendency of neurotic consumers to buy cryptocurrency and does the same for agreeable consumers. In contrast, FOMO as a fear inducer is less effective in increasing cryptocurrency purchase intention for conscientious and open consumers. The major takeaway is the recognition that while FOMO can be a powerful motivator for consumers' cryptocurrency consumption decisions, its effect is not equal for all big five personality traits.

Additional insights are discovered, particularly as it relates to consumers higher on openness to experience scores. Contradicting our proposed hypothesis, which was grounded in existing literary support (see Brown & Taylor, 2014), the present study's results indicate that openness to experience did not directly affect cryptocurrency purchase intentions. But when it did, this impact was negatively mediated as a result of FOMO. For decades, researchers have concluded that highly open consumers are often the first group of people to try a new innovation (see Digman, 1990). These are people willing to take greater risks and take on new ideas (McCrae & Costa, 1997). Therefore, it is striking that openness to experience does not affect cryptocurrency purchase intention and when it does through the mediating role of FOMO, higher openness values decrease consumers purchase intention. These results contrast the findings of Brown and Taylor (2014) who found that this open trait was associated with increased stock market participation. Although both stock investing and cryptocurrency investing are risky, the findings clearly demonstrate that open consumers appear to be wearier of cryptocurrencies since cryptocurrencies are generally more volatile than stocks (Royal, 2023). As such, we urge academic researchers not to treat cryptocurrency investing and stock investing similarly nor to equate their consuming segments as the same. Considering these findings, it is feasible to conclude that the psychological motivators for cryptocurrency consumption are different from stock market participation.

Additional significance of this research lies in our investigation of cryptocurrency knowledge. The findings highlight that cryptocurrency knowledge does moderate the relationship between personality traits and FOMO, which ultimately can affect cryptocurrency purchase intention. Interestingly, we found that cryptocurrency knowledge decreased FOMO for extraversion, openness to experience, and conscientiousness, while increasing FOMO for consumers with higher neuroticism traits. These findings suggest that knowledge plays a conditional role in shaping the relationship between personality traits and cryptocurrency investing behavior. In fact, we show the importance of considering personality traits and cryptocurrency knowledge as influential factors for understanding consumer behavior in the cryptocurrency market. As each consumer possesses a unique combination of the traits that characterizes their personality, we find that knowledge affects consumers differently. For some personality traits (agreeableness and neuroticism), more cryptocurrency knowledge has a positive influence on their decision to invest, while for others (conscientious and openness to experience) more cryptocurrency knowledge further dissuades their consumption decision in a bear market. In a bullish market, more cryptocurrency knowledge translated into purchase indifference for highly conscientious and open consumers, which indicated their need for greater market certainty.

According to prospect theory, consumer's decisions are influenced by their perception of potential gains and losses (Kahneman & Tversky, 1979). Our findings are consistent with this theory as we found that consumers with traits such as conscientiousness and openness to experience consider cryptocurrency as a perceived potential loss as these traits showed negative mediating effect with FOMO in a bear market. In contrast, neurotic and agreeable consumers considered it a worthwhile purchase (highlighting perceived gains) as these consumers showed positive mediating effect with FOMO. Moreover, we observed that higher cryptocurrency knowledge increased FOMO for traits such as neuroticism and agreeableness, further emphasizing the perceived gains aspect of prospect theory.

Our study also revealed that neurotic and agreeable consumers were not interested in purchasing cryptocurrency. However, with FOMO and knowledge, these consumers shifted their reference points toward perceived gains as indicated by prospect theory. Similarly, conscientious consumers were interested in purchasing cryptocurrency without FOMO. However, with FOMO and knowledge, these consumers shifted their reference points toward perceived losses, showing lesser intention to purchase cryptocurrencies. Therefore, by utilizing prospect theory, we were able to demonstrate how FOMO can lead to perceived losses and risk-aversion for some personality traits, while for others, it can lead to perceive gains and risk-seeking behavior, particularly in a bear market.

Overall, the findings of this study contribute to the growing body of academic literature on cryptocurrency purchase intentions by providing insights into the complex interplay between personality traits, emotions (FOMO), and knowledge in shaping consumers' consumption decisions. It is important to consider that consumers may invest in cryptocurrencies for reasons such as diversification, hedging against inflation, or seeking returns. But as we find, consumers may also invest in cryptocurrencies as a result of their personality traits, FOMO, and knowledge. Clearly young adults have limited knowledge, and with the internet and online retail trading communities they are able to access misinformation about cryptocurrency. Therefore, academic researchers exploring cryptocurrency are encouraged to take a 360° view of the domain, particularly concerning the role of knowledge to further comprehend factors that influence cryptocurrency consumption.

5.12 | Managerial implications

In concurrence with Anaza et al. (2022), we believe that cryptocurrencies as a form of digital payments should not only be viewed from an economic or financial perspective, but also as a marketing opportunity. Cryptocurrency presents an exciting opportunity for businesses to reach

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new customers and markets that would otherwise be difficult to reach. Thus, our goal with this research is to increase the accurate psychographic segmentation and targeting of cryptocurrency consumers using personality traits and emotional appeals.

This research has several managerial implications that may help marketing managers understand and segment consumers while increasing consumers movement through each marketing funnel stage. Research and practice within branding posit that the creation of different brands with varying brand personalities at contrasting price points may successfully attract different market segments by employing various brand communication strategies (Ang & Lim, 2006; Batra et al., 2010). Of importance, our research indicates that consumer groups with varying personality traits hold different levels of intention to purchase cryptocurrency, which is mediated by FOMO and moderated (at varying levels) by cryptocurrency knowledge. Our findings suggest that these personality traits can be an accurate basis for market segmentation. To this end, our research indicates that marketing managers may profit from creating different brands of cryptocurrency that target each personality trait as a separate segment. Here, we suggest that managers develop several branded offerings, each with a distinct brand personality, communication strategy, and price point that speaks best to their targeted personality segment(s). Further, marketing managers for organizations that seek to educate consumers on the dangers of cryptocurrency should seek to educate consumers in a manner that decreases FOMO and thus mitigates its effect.

The finding of this study suggests that consumers with different personality traits hold different levels of purchase intention for cryptocurrency. Further, this research offers what role FOMO and knowledge of cryptocurrency plays in this relationship. Our findings suggest that consumers with different personality traits would respond differently to advertising strategies, particularly those using a FOMO advertising appeal. These findings hold vital information for marketing managers. For example, our research indicates that FOMO most influences consumers with agreeable and neurotic personality traits. As fear appeals are one of several promotional strategies employed by marketers, marketing communications that employ a fear appeal to elevate FOMO would prove advantageous in moving these consumers from awareness to consideration to purchase.

Additionally, given the mixed effects of knowledge on FOMO, managers should alter their educational approach on cryptocurrency for certain groups of consumers (conscientious and open to experience consumers) and build brands by strengthening their emotional appeals. This has become the staple in other industries. For example, within the insurance industry, All State's "Avoid Mayhem" commercials, while funny, employ a fear appeal, and both Geico and Progressive leverage fun appeals.

5.13 | Limitations and future research opportunities

One limitation of the research is that knowledge, as defined here, is measured in terms of "more or less," with no indication concerning whether the amount of knowledge is "favorable or unfavorable." Indeed, the direction of the knowledge should correlate to the positive or negative attitudes, or beliefs respondents would hold about cryptocurrency. Since study one was conducted at a time when the whole cryptocurrency market was in a bear market, it can be predicted that much respondents' knowledge included the recent decline of cryptocurrency's value. Knowledge can help form positive or negative evaluations, and the current information or most salient and recent experiences with cryptocurrency during this study were likely to be negative. This knowledge could counter FOMO as it indicates that there is nothing to fear missing out on. Future research should build upon this work by measuring if knowledge about cryptocurrency has created negative or positive attitudes or beliefs about cryptocurrency. While this study was conducted in the United States (the world's largest cryptocurrency market), the generalizability of our findings could be expanded to identify the role that personality traits and FOMO play on consumers' purchase intentions of cryptocurrencies within different domestic and foreign cultural groups, geographic locations, and genders.

In addition to demographic and risk tolerance factors, cryptocurrency purchase intentions can likely be influenced by other variables such as market trends, media exposure, regulatory environment, and personal financial literacy. Although not the focus of this study, these macroenvironmental factors may explain the nonpersonality drivers of FOMO within a cryptocurrency context. Therefore, future studies will benefit from exploring the macroenvironmental predictors of cryptocurrency consumption. In a similar regard, the regulatory environment surrounding cryptocurrency marketing may also affect consumers behavior. For example, in the United States, the US Securities and Exchange Commission (SEC) has expressed concerns about the need to regulate cryptocurrencies (Duggan & Adams, 2023), which can impact consumers' ability to purchase and trade cryptocurrencies in the near future. Different countries have varying approaches to cryptocurrency regulation, with some, like China, implementing bans on cryptocurrencies (Shin, 2022). These external factors which are beyond the control of individual consumers significantly influence the dynamics of the cryptocurrency market and can contribute to consumers' FOMO. As researchers, we acknowledge that these external factors present research opportunities to further advance our understanding of cryptocurrency marketing. Longitudinal panel data was sort from study 1 participants, a year later, but due to a low response rate (n = 50), the researchers decided to forfeit this approach in favor of a repeated cross-sectional sampling method with completely different participants. Without the challenges of panel data collection, future research studies will benefit immensely from following-up with participants over time, particularly given the volatility in the cryptocurrency market.

Furthermore, it is imperative to exercise caution when interpreting the results of study 2, as the data set utilized does not comprehensively embody the characteristics of a bull market within the cryptocurrency domain. This is due to the persistent presence of potential market volatility, stemming from regulatory uncertainties that do not fully align with the sentiments of cryptocurrency

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stakeholders, including investors and consumers who maintain a vested interest in this domain. Consequently, we strongly advocate for future researchers to undertake a more robust investigation during a distinct bull market, exemplified, for instance, by the cryptocurrency market conditions in 2021, marked by Bitcoin's price ranging between $33,114.36 and $46,306.45 (Yahoo Finance, 2023), a period widely regarded as an epic bull market. It is posited that the personality trait distinctions observed during such a period would more accurately encapsulate the targeted observations elucidated in study 2.

ACKNOWLEDGMENTS

Financial support for this research was provided through the 2023 College of Business & Analytics Summer Faculty Research Grant at Southern Illinois University Carbondale.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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How to cite this article: Anaza, N. A., Upadhyaya, B., Bennett, D., & Ruvalcaba, C. (2024). Is it FOMO or is it ME? The influence of personality traits on cryptocurrency consumption. *Psychology & Marketing*, *41*, 184–202. https://doi.org/10.1002/mar.21919

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